

The Library of Life

II PUC · Ch 5 · Molecular Basis of Inheritance

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



The Secret Book

"Every cell keeps a secret book — **DNA**," whispered Prof. BAIO. In 1953, **Watson and Crick** showed it is a **double helix**: two antiparallel strands, bases paired A with T and G with C (**Chargaff's rule**). The steps are 0.34 nm apart and one turn is 3.4 nm.

💡 **REMEMBER**

A = T by 2 hydrogen bonds · G ≡ C by 3 hydrogen bonds.



Packing the Book

Human DNA is about 2.2 metres long, yet it fits in a tiny nucleus! It wraps around eight **histone** proteins — a **histone octamer** — to form a **nucleosome** of about 200 base pairs. Chromatin looks like beads on a string.

> 💡 **REMEMBER**
> Loosely packed, active chromatin = euchromatin · densely packed, inactive = heterochromatin.



Which Is the Real Book?

Griffith (1928) found a "transforming principle" in bacteria. Then **Hershey and Chase** (1952) labelled viruses with radioactive ^{32}P (DNA) and ^{35}S (protein). Only the ^{32}P went inside the bacteria — so DNA is the genetic material!

💡 **REMEMBER**

RNA is the genetic material in some viruses; DNA is more stable than RNA.



The Photocopier

Before a cell divides, DNA copies itself — **replication**. Meselson and Stahl (1958) showed it is **semi-conservative**: each new DNA keeps one old strand. **DNA polymerase** builds only 5′ → 3′, so the **leading strand** is continuous and the **lagging strand** is made in **Okazaki fragments**, joined by **DNA ligase**.

💡 **REMEMBER**

Replication starts at a definite point — the origin of replication.



Copying One Page

To use one gene, the cell copies it into RNA — **transcription**. RNA polymerase binds the **promoter**, copies the structural gene and stops at the **terminator**. In eukaryotes the first copy, **hnRNA**, is edited: **splicing** removes introns and joins exons, then capping and tailing finish it.

💡 **REMEMBER**

Transcription unit = promoter + structural gene + terminator.



The Code

The **genetic code** is read in triplets — **codons**. 61 codons stand for amino acids and 3 are stop codons (UAA, UAG, UGA). **AUG** is the start codon for methionine. The code is degenerate and nearly universal. **tRNA** reads codons with its anticodon — the adapter molecule.

💡 **REMEMBER**

There are 64 codons in all.



Building the Protein

On the ribosome, **translation** begins at AUG. tRNAs, charged with amino acids, bring them one by one; peptide bonds join them during elongation. At a stop codon, a release factor ends the chain — a new protein is born!

💡 **REMEMBER**

The ribosome is also an enzyme: its 23S rRNA (in bacteria) is a ribozyme.



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Switches and Fingerprints

Genes have switches. In the **lac operon** (Jacob and Monod), lactose is the inducer — it removes the repressor so the genes switch on. The **Human Genome Project** read about 3 billion base pairs (~20,500 genes). **DNA fingerprinting** uses repeated DNA (**VNTRs**) to identify people.

💡 REMEMBER

DNA fingerprinting was developed by Alec Jeffreys.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

DNA p.1 · Watson and Crick p.1 · double helix p.1 · Chargaff's rule p.1 · histone p.2 · histone octamer p.2 · nucleosome p.2 · Hershey and Chase p.3 · replication p.4 · semi-conservative p.4 · DNA polymerase p.4 · leading strand p.4 · lagging strand p.4 · Okazaki fragments p.4 · DNA ligase p.4 · transcription p.5 · promoter p.5 · terminator p.5 · hnRNA p.5 · splicing p.5 · genetic code p.6 · codons p.6 · AUG p.6 · tRNA p.6 · translation p.7 · lac operon p.8 · Human Genome Project p.8 · DNA fingerprinting p.8 · VNTRs p.8



QUICK QUIZ

Do you remember?

1. Who proposed the double helix model?

- Hershey and Chase
- Watson and Crick
- Meselson and Stahl
- Jacob and Monod

2. DNA replication is:

- Conservative
- Semi-conservative
- Dispersive
- Random

3. Okazaki fragments form on the:

- Leading strand
- Lagging strand
- Template only
- mRNA

4. The start codon is:

- UAA
- AUG
- UGA
- UAG

5. The inducer of the lac operon is:

- Glucose
- Lactose
- Galactose
- Repressor

Answers: 1-Watson and Crick · 2-Semi-conservative · 3-Lagging strand · 4-AUG · 5-Lactose